

# Quantitative Spatial Models for Transport Appraisal

Optional capstone add-on: Berlin's U5 connection

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Problem and solution manual

# Policy question

- ▶ How should we **appraise a transport investment when households, firms, wages, rents, employment, and location choices respond?**
  - ▶ Conventional appraisal emphasizes time savings and user benefits.
  - ▶ A quantitative spatial model **adds endogenous sorting and general-equilibrium adjustment** to conventional appraisal.
- ▶ Highly topical
  - ▶ Allen, Fuchs, and Wong, *From Social Savings to Spatial Equilibrium*, RSUE (forthcoming) for further reading.
  - ▶ Currently working on related implementation work with the UK Department for Transport (with Sidharth Moktan, and Daniel Sturm).
- ▶ This capstone applies that logic to a **concrete case combining the ARSW2015 toolkit and the TTMATRIX toolkit.**

# A quantitative transport-appraisal workflow

1. Define two comparable transport networks.
  - ▶ Baseline network
  - ▶ Counterfactual policy network (usually with extension)
2. Compute bilateral travel-time matrices.
  - ▶ Map travel times into commuting costs (usually via iceberg cost in model).
3. Quantify a baseline spatial equilibrium.
  - ▶ Estimate or set all parameters
  - ▶ Invert fundamentals conditional on parameter values and observed data
4. Solve the counterfactual policy equilibrium.
  - ▶ Compute relative changes from baseline to counterfactual equilibrium,  $\hat{x} = \frac{x'}{x}$
  - ▶ Compute local changes in prices and quantities
  - ▶ Compute aggregate changes in welfare and population.

# Running example: Berlin's central U5 connection



Stylised network comparison for quantitative transport appraisal

- ▶ Berlin's most recent U-Bahn extensions created a new east–west connection through the historic centre:
  - ▶ **2009:** Hauptbahnhof–Brandenburger Tor opened as the U55.
  - ▶ **2020:** Brandenburger Tor–Alexanderplatz opened, closing the gap and integrating the U55 into the U5.
- ▶ Our counterfactual adds **both new sections** to generate a sizable accessibility shocks

# Network construction

## Baseline network (2005 perspective)

- ▶ Take current Berlin line and station layers
- ▶ Delete Alexanderplatz–Hauptbahnhof connection
  - ▶ Copy the line layer in e.g. QGIS
  - ▶ Delete all corridor segments between the endpoints
  - ▶ Preserve intersecting services
  - ▶ Treat observed stations consistently

## Counterfactual network

- ▶ Current network  $\equiv$  future network from 2005 perspective
- ▶ Counterfactual network differs from baseline network by adding U5 Alexanderplatz–Hauptbahnhof extension

## From GIS to bilateral travel times

- ▶ For transport counterfactuals, **need to compute travel times them ourselves**
  - ▶ Historic or future travel times can typically not be queried
  - ▶ Important that we use **exactly the same modelling and parameters**
- ▶ Travel times in ARSW2015 toolkit are fairly sophisticated (took us ages...)
  - ▶ U-Bahn, S-Bahn, Tram, Bus, interchange penalties, etc.
- ▶ **TTMATRIX toolkit designed for quick stylized counterfactuals.**
  - ▶ Just needs origins-destination shapefile, one network shapefile, 2 speed parameters
    - ▶ Station shapefile is optional (here, use is highly recommended).
  - ▶ Same blocks as in ARSW2015 toolkit; default speed parameters applicable.
  - ▶ Line shape corresponds to *counterfactual*. You just need to create *baseline network*.
  - ▶ **Run TTMATRIX toolkit for *counterfactual* and baseline *baseline network***
  - ▶ Generate two travel time matrices.
    - ▶ **Only change the network**, keep everything else (including parameters) the same.

## Figure 2: direct travel-time effects

### Direct accessibility effects of the central U5 connection

A. Mean travel time



B. Blocks reachable within 30 minutes



Relative changes = 100 x (with U5 - without U5) / without U5. -11: The origin block itself is excluded; red line shows the U5 connection.

# Matrix validation before economics

- ▶ **Same dimension and exact identifier ordering.**
  - ▶ Minutes, not seconds; documented diagonal convention.
  - ▶ No unexplained missing or infinite OD pairs.
  - ▶ Residence-by-workplace orientation consistent with model.
    - ▶ Here commuting costs are symmetric, so does not matter
- ▶ **Accessibility indices should make sense**
  - ▶ Higher accessibility in centre and near stations
  - ▶ For affected OD pairs, complete-network travel times should weakly improve relative to the reduced network.
  - ▶ There should be no sizable travel time increases.
  - ▶ Check (again) after reading into MATLAB (using MAPIT)

# Bridge to the ARSW model

Let  $t_{ij}$  denote travel time from residence  $i$  to workplace  $j$ .

1. Ensure both TTMATRIX outputs and ARSW-toolkit block index are in same order.
  - ▶ Ensured by construction, toolkits are compatible ✓
2. Convert  $t_{ij}$  into the commuting-cost object used by the model.
  - ▶ Ensured via exponential iceberg transport cost ✓
3. Treat the reduced network as the baseline transport system.
  - ▶ Replace the original ARSW-toolkit with the baseline TTMATRIX.
4. Recover or re-invert baseline fundamentals under that system.
  - ▶ Keep the parameter values constant at values used/estimated by ARSW.
5. Load the ARSW-toolkit counterfactual matrix and solve the model
  - ▶ While holding the chosen fundamentals and parameters fixed.

# Equilibrium outcomes

## ► Before starting with the simulations

- **Inspect the fundamentals** inverted under the new travel times
- Commuting market access is particularly relevant
- All looks good  $\Rightarrow$  most likely no ordering issue with blocks ✓

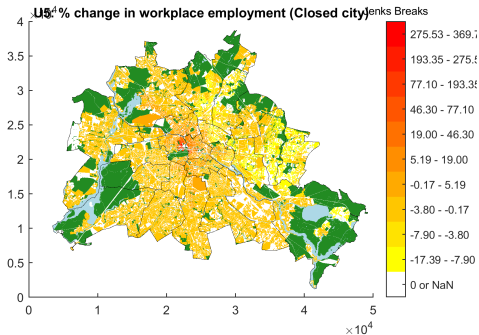
Inverted primitives

## ► Ready to simulate the model in the open-city and closed-city scenarios

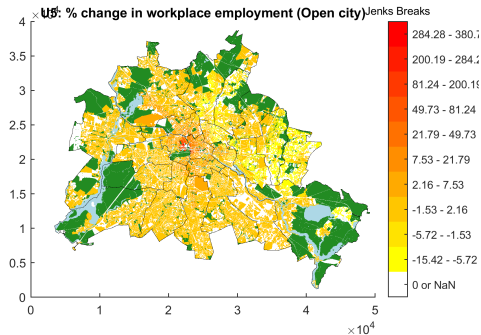
- Residence and workplace employment
- Wages and expected worker income
- Floor-space prices and development density
- Commuting flows and average commuting costs
- Productivity and amenity feedbacks under endogenous agglomeration
- Welfare or aggregate population

# Employment response: closed versus open city

## Closed city

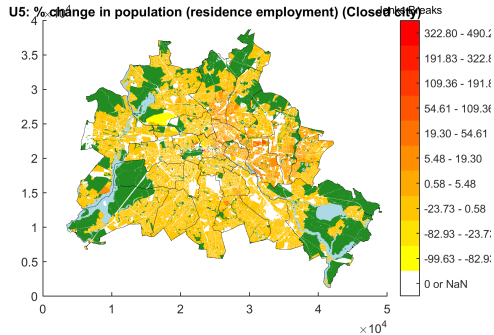


## Open city

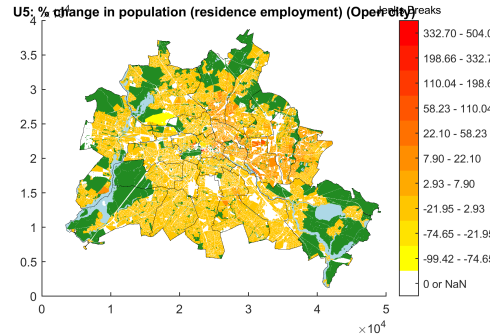


# Population response: closed versus open city

## Closed city

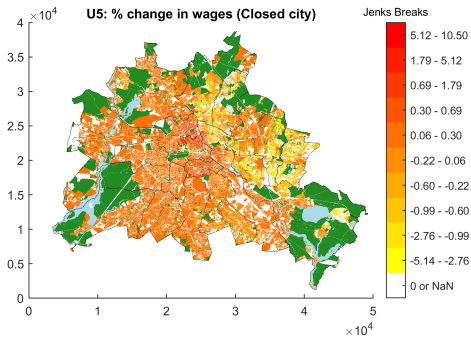


## Open city

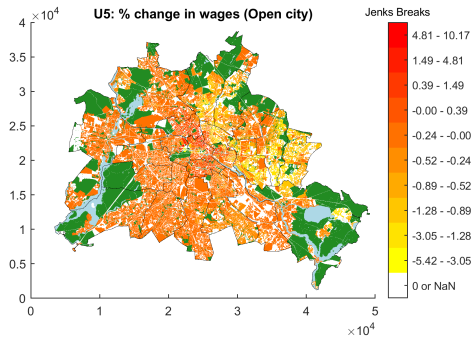


# Wage response: closed versus open city

## Closed city

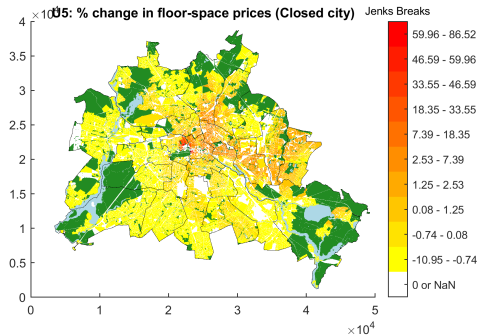


## Open city

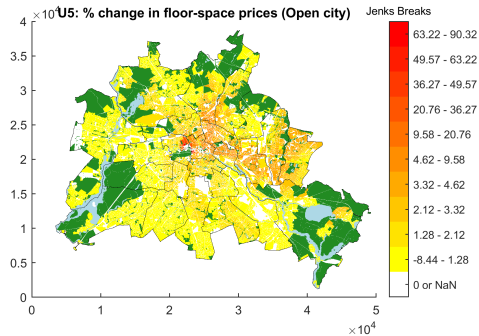


# Floor-space price response: closed versus open city

## Closed city



## Open city



## Aggregate effects depend on the spatial closure

| Closure     | Expected utility | Population | GDP     |
|-------------|------------------|------------|---------|
| Closed city | +0.444%          | 0.000%     | +0.307% |
| Open city   | 0.000%           | +2.340%    | +2.348% |

- ▶ In the closed city, the transport improvement raises welfare because total population is fixed.
- ▶ In the open city, expected utility is pinned down by the outside option; the adjustment occurs through migration.

# A rough annual monetary equivalent of the welfare gain

## Back-of-the-envelope scaling

$$0.4439\% \times \underbrace{\text{EUR 37.347 bn}}_{\substack{\text{Berlin gross wage bill, 2006} \\ \text{excluding marginal employment}}} \simeq \text{EUR 166 million per year.}$$

- ▶ This computes the monetary equivalent of the closed-city welfare effect.
  - ▶ Recall that welfare scales linearly in wage
  - ▶ A recurring **annual** equivalent under the model's new spatial equilibrium.
- ▶ It does not yet deduct construction, operating or maintenance costs.
  - ▶ Notice that operating costs are about covered by fees

Source: Amt fuer Statistik Berlin-Brandenburg, VGR, P I 2, table 18. Model outcomes: authors' calculations.

## Benefits and costs on a common 2006 basis

$$PV_{2006}(B) = \frac{\text{EUR } 166\text{m}}{0.05} \simeq \text{EUR } 3.32\text{bn.}$$

- ▶ Deflate both segments with the Destatis price index for engineering structures:
  - ▶ U55: EUR 320m in 2009 → EUR 288m in 2006 prices.
  - ▶ U5 gap: EUR 525m at the 2013 price level → EUR 443m in 2006 prices.
- ▶ Combined construction cost:  $C_{2006} \simeq$  **EUR 731m.**

$$\frac{PV_{2006}(B)}{C_{2006}} = \frac{\text{EUR } 3.32\text{bn}}{\text{EUR } 0.731\text{bn}} \simeq \mathbf{4.5}.$$

- ▶ High benefit-cost ratio, but stylized modelling of travel times **likely overstates benefits**

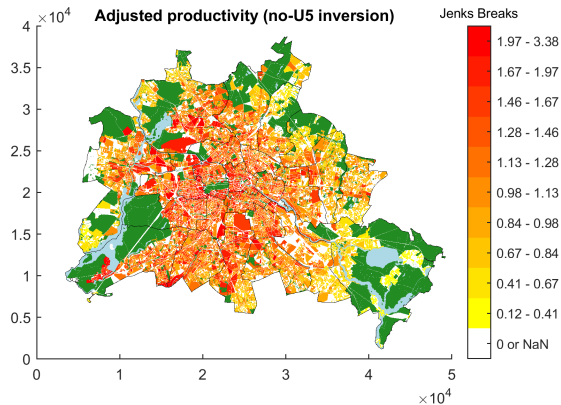
Sources: BVG; Die Zeit; Destatis. Reported totals are price-deflated because annual expenditure profiles are unavailable.

# Capstone task

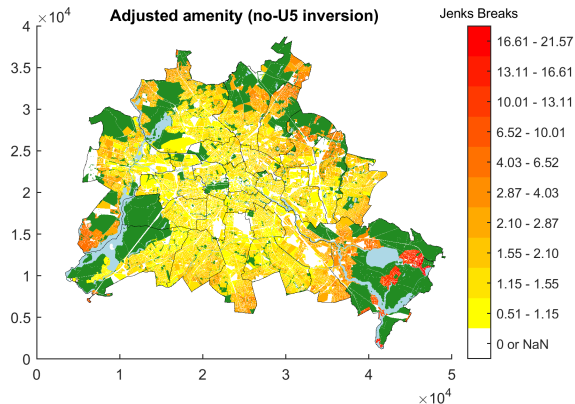
1. Produce and audit the two network scenarios.
2. Compute and validate both travel-time matrices.
3. Quantify model.
4. Solve the complete-network counterfactual and do benefit-cost calculation.
5. Explain mechanisms, distributional patterns, and limitations.

**Nerd task:** Use more sophisticated travel time matrices incorporating additional public and private modes...

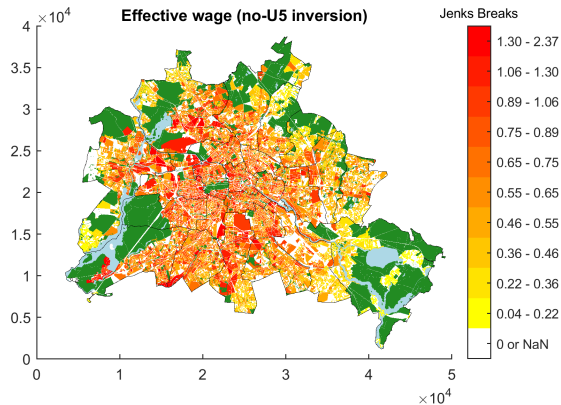
# Inverted adjusted productivity

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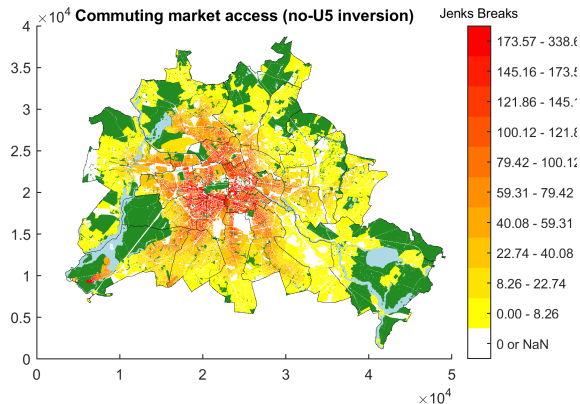
# Inverted adjusted amenity

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# Inverted effective wage

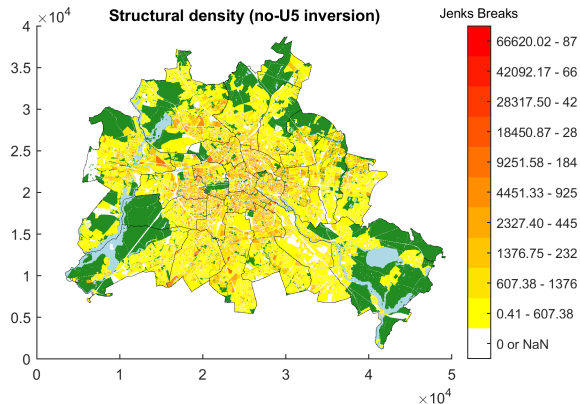
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# Inverted commuting market access



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# Inverted structural density

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